Quantum Happiness
Quantum Poetry
Quantum Teaching
Quantum Faith
Quantum Love

For a branch of physics that is often described as only being relevant to microscopic processes involving subatomic particles, that's a pretty impressive résumé.

To be fair, quantum mechanics—or "quantum physics," or "quantum theory," the labels are all interchangeable—is not only relevant to microscopic processes. It describes the whole world, from you and me to stars and galaxies, from the centers of black holes to the beginning of the universe. But it is only when we look at the world in extreme close-up that the apparent weirdness of quantum phenomena becomes unavoidable.

One of the themes in this book is that quantum mechanics doesn't deserve the connotation of spookiness, in the sense of some ineffable mystery that it is beyond the human mind to comprehend. Quantum mechanics is *amazing*; it is novel, profound, mind-stretching, and a very different view of reality from what we're used to. Science is like that sometimes. But if the subject seems difficult or puzzling, the scientific response is to solve the puzzle, not to pretend it's not there. There's every reason to think we can do that for quantum mechanics just like any other physical theory.

Many presentations of quantum mechanics follow a typical pattern. First, they point to some counterintuitive quantum phenomenon. Next, they express bafflement that the world can possibly be that way, and despair of it making sense. Finally (if you're lucky), they attempt some sort of explanation.

Our theme is prizing clarity over mystery, so I don't want to adopt that strategy. I want to present quantum mechanics in a way that will make it maximally understandable right from the start. It will still seem strange, but that's the nature of the beast. What it won't seem, hopefully, is inexplicable or unintelligible.

We will make no effort to follow historical order. In this chapter we'll look at the basic experimental facts that force quantum mechanics upon us, and in the next we'll quickly sketch the Many-Worlds approach to making